Laiba Fatima
22K-5195
BSE-5B

PAGE
DATE

DB A2

Q1. a) Entitities : Patient , Treatment , Procedure , Medication

Attributes:

PatientID	PatientName

TreatmentID	PatientID	HospitalName	StartDate	EndDate

ProcedureID	Treatment ID	Procedure Name	Date

MedicationID	TreatmentID	MedicationName	Dosage	Duration

Datatypes:

1. PatientID Number Primary key
2. PatientName varchar (30)

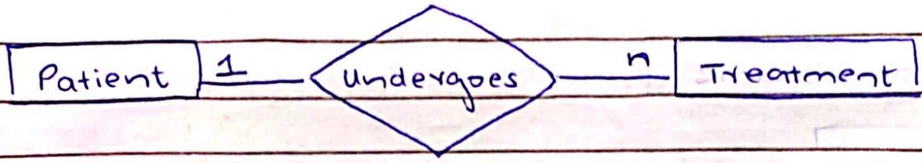
1. TreatmentID Number Primary key
2. Patient ID Number Foreign key (PatientID) references Patient (PatientID)
3. HospitalName varchar (100)
4. StartDate date
5. EndDate date

1. ProcedureID Number Primary Key
2. Treatment ID Number Foreign key (Treatment ID) references Treatment (TreatmentID)
3. ProcedureName varchar (50)
4. Date date

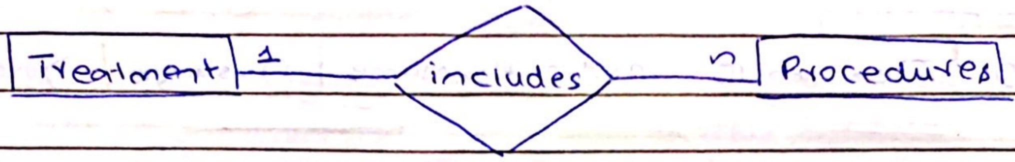
1. MedicationID Number Primary key
2. Treatment ID Number Foreign key (TreatmentID) references Treatment (TreatmentID)
3. Medication Name varchar (30)
4. Dosage Number
5. Duration Number

PAGE
DATE

Relationships:



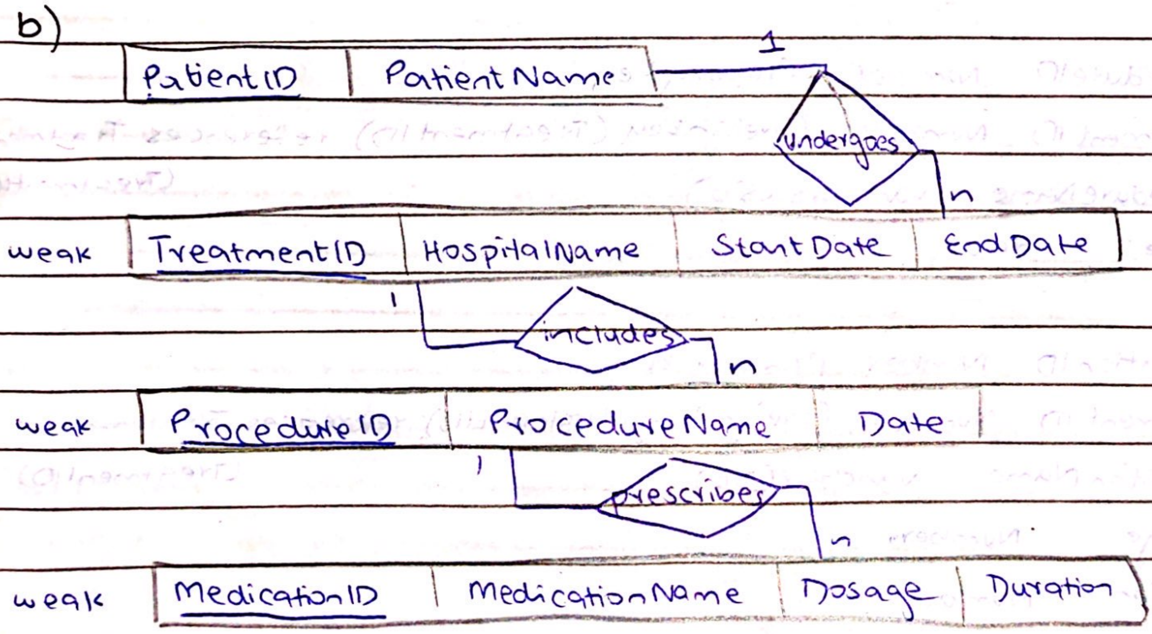
One Patient can undergo many treatments (1 to many)



One treatment can include multiple procedures (1 to many)

One treatment can prescribe many medications (1 to many)

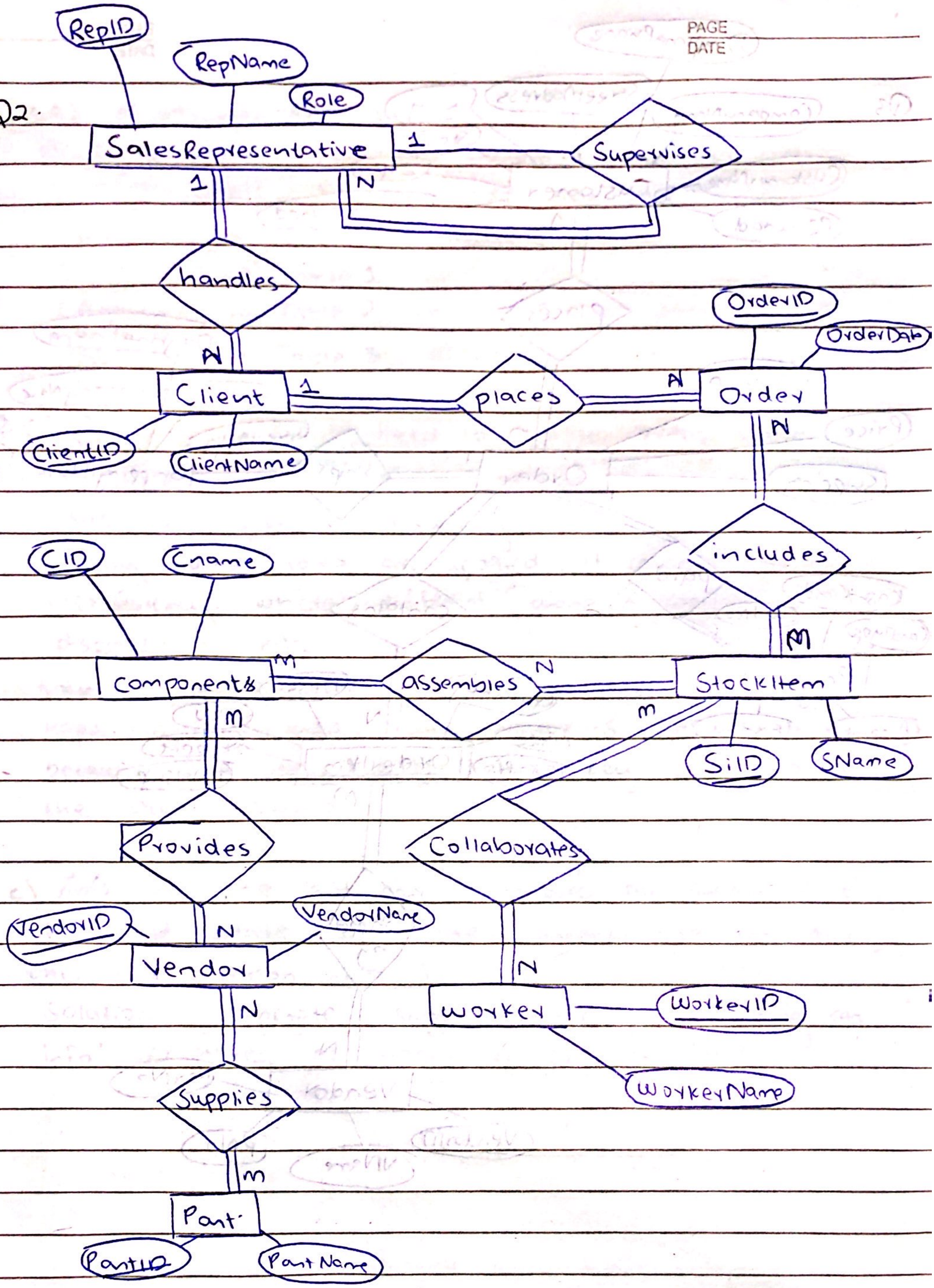
b)



- No foreign keys

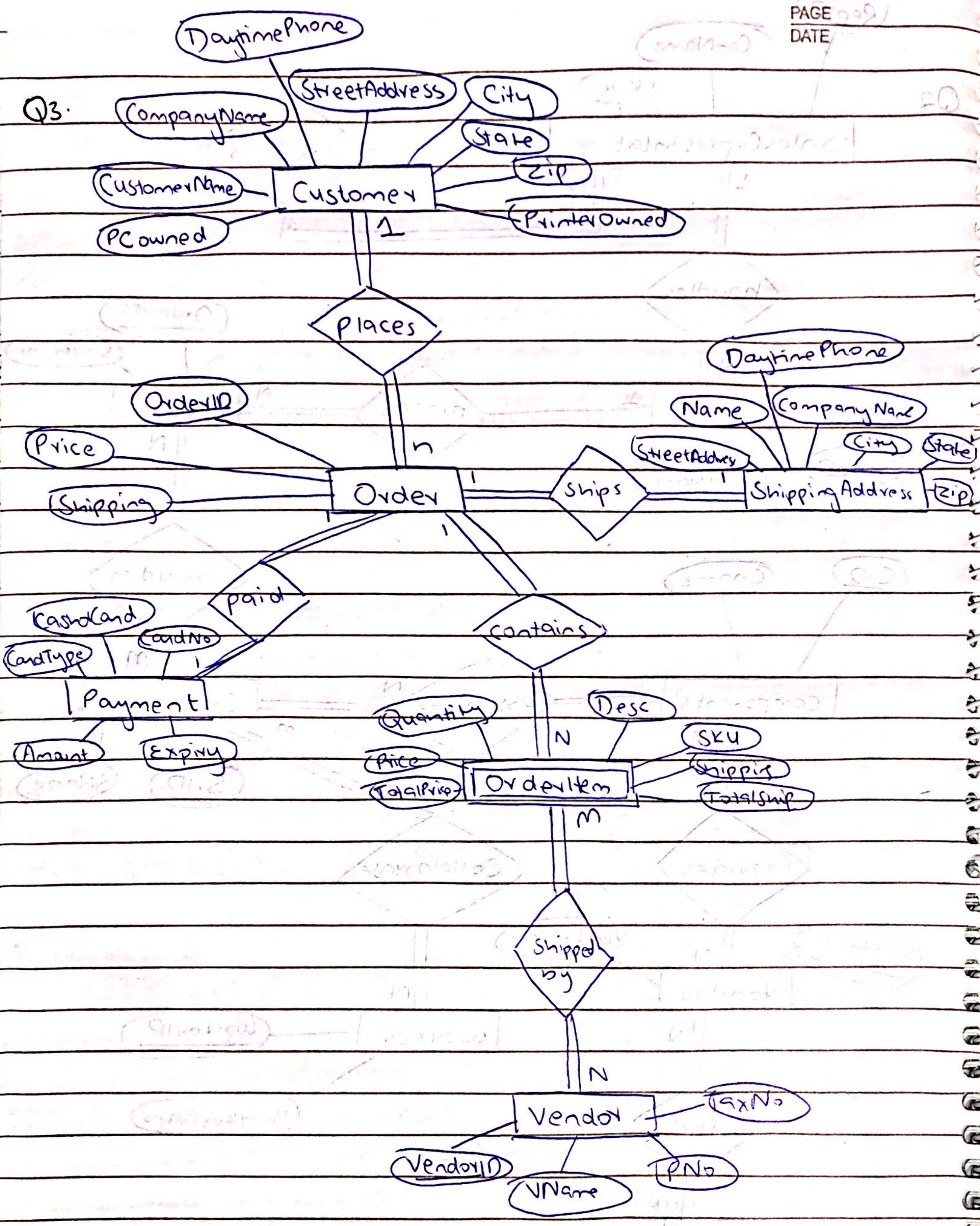
PAGE
DATE

Q2.


RepID
RepName
Role
SalesRepresentative
1
Supervises
1
N
handles
N
Client
1
places
N
Order
OrderID
OrderDate
ClientID
ClientName
N
includes
M
CID
Cname
components
m
assembles
N
StockItem
m
SiID
SName
m
Provides
Collaborates
VendorID
VendorName
N
Vendor
N
N
worker
WorkerID
WorkerName
Supplies
m
Part
PartID
Part Name

PAGE
DATE

Q3.



DaytimePhone
CompanyName
StreetAddress
City
State
Zip
CustomerName
Customer
PrinterOwned
PCowned
1
places
n
OrderID
Price
Shipping
Order
1
ships
1
DaytimePhone
Name
CompanyName
StreetAddress
City
State
ShippingAddress
Zip
1
paid
1
CashOrCard
CardNo
CardType
Payment
Amount
Expiry
1
contains
N
Quantity
Desc
SKU
Price
Shipping
TotalPrice
OrderItem
TotalShip
M
shipped by
N
Vendor
TaxNo
VendorID
VName
PNo

PAGE
DATE

Q4 a) ~~A~~ Insertion Anomaly
CarID is primary key and therefore can't be null in current table so can't add new supplier
Solution: Divide data into multiple tables
Cars → Table 1
Category → Table 2
Supplier → Table 3

b) Current table would need to update every single row eventhough its just updation for 'Hertz', this is updation anomaly.
if only some rows are updated, it causes data inconsistency where different phone numbers are displayed for Hertz
Solution: Supplier info in supplier table so updating happens only once, which maintains data consistency because you can assign foreign key SupplierID to the other tables.

c) ~~This~~ If the last car is removed for supplier 'Avis', it would delete the entire supplier info for 'Avis', this is deletion anomaly.
Solution: seperate supplier table so deleting car info will have no impact on supplier table.

PAGE
DATE

Q4b) no multivalued attributes

Cars — Category — Supplier

1NF : Primarykeys : CarID , CategoryID , SupplierID

2NF :

Cars — Foreign key: CategoryID , SupplierID

CarID	Model	PricePerDay	CategoryID	SupplierID
C101	Toyota Camri	5000	CT1	S1
C102	Honda Civic	4500	CT1	S2
C103	Ford Explorer	7000	CT2	S1
C104	Chevrol et Tahoe	8000	CT2	S3
C105	Toyota Corolla	4800	CT1	S1
C106	Tesla Model X	9500	CT3	S4
C107	Jeep Cherokee	7500	CT2	S2
C108	Nissan Leaf	9000	CT3	S4

~~2NF~~ Category

CategoryID	CategoryName
CT1	Sedan
CT2	SUV
CT3	Electric

Supplier

SupplierID	SupplierName	Email	Phone
S1	Hertz	hertz@cars.com	123-456-789
S2	Enterprise	enterprise@cars.com	987-654-321
S3	Avis	avis@cars.com	654-321-987
S4	Sixt	sixt@cars.com	741-852-963

- already in 3NF : no transitive dependency

~~3NF~~

- No Insertion Anomaly: You can add new category, supplier irrespective if we don't have car

- No updation Anomaly: Phone number for supplier updates only once maintaining data consistency

- No deletion anomaly: deleting car has no effect on category or supplier

Advantages
- data consistency
- data integrity: foreign key

Q5.

1. Functional Dependencies
Patient_Number → FullName, BedNumber, wardNumber, ward Name

DrugNumber → Name, Description, Dosage

PatientNumber, DrugNumber → Method of Admin, Unitsperday, Start Date, FinishDate

~~2. already in 1NF~~

~~2,~~ 2NF

PatientNumber	FullName	BedNumber	WardNumber	WardName

DrugNumber	Name	Description	Dosage

PatientNumber	DrugNumber	MethodofAdmin	Units	Start	Finish

3NF : no transitive dependency

PK & FK identified in diagram

Q6.

Candidate key: ReservationID + CustomerID

Dependency: RoomType on RoomNumber } partial dependency
ServiceCharge on ServiceCode

Not in 2NF

Decompose:

CustomerID	ReservationID	RoomNumber	StayDate	ServiceCode

RoomNumber	RoomType

ServiceCode	ServiceCharge

- No need for 3NF because no transitive dependency